

JIAYAN WEN

Low-Level Software & Computer Systems | C/C++ | Embedded Systems | Performance
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EDUCATION

University of Pennsylvania | M.S.E. Electrical and Computer Engineering Philadelphia, PA | Incoming Fall 2027
Santa Clara University | B.S. Electrical and Computer Engineering Santa Clara, CA | 2023–2026
Relevant Coursework: Operating Systems, Computer Architecture, Embedded Systems, Microprocessor Systems, Digital Logic Design, Digital Signal Processing

TECHNICAL SKILLS

Languages: C, C++, Python, SystemVerilog, ARM Assembly
Systems & Embedded: GNU/Linux, STM32 Cortex-M, USB Host/FATFS, SPI, interrupts, peripheral drivers, memory-mapped and CPU-visible interfaces
Performance & Debugging: perf, hardware counters, cache/SIMD/vectorization analysis, reproducible benchmarks, reference models, failure isolation
Tools: Git, CMake, Ninja, GCC/G++, Verilator, Vivado, MATLAB

SELECTED LOW-LEVEL PROJECTS

CPU Matrix-Multiplication Performance Study | C++, perf Independent Project | 2026–Present

- Built FP32 i-j-k, i-k-j, and transposed-B kernels with a strict reference model and reproducible benchmark; isolated the N=256 throughput collapse through controlled sweeps and perf, where L1D miss rate rose from about 7% to 81.94%.
- Used GCC vectorization reports, hardware counters, and assembly inspection to separate memory-layout effects from arithmetic semantics; kept AVX2 reassociation in a separate fast mode after detecting changed cancellation behavior.

STM32 Embedded Firmware and Hardware Interfaces | C, ARM Assembly Course Project

- Developed STM32L476 firmware integrating USB Host/FATFS, DAC, SPI, interrupts, and peripheral drivers; debugged register-level state and firmware/hardware interactions using C and ARM assembly.

Output-Stationary Systolic Array | SystemVerilog, Verilator Independent Project | 2026–Present

- Designed a parameterized accelerator with independent array and inner-product dimensions, a skewed feeder, distributed MAC accumulation, and cycle-accurate start/busy/done control.
- Built automated Verilator regression across eight configurations and 872 end-to-end operations, with reference-model, pairing, MAC-count, and protocol monitors; maintained warning-clean builds and reproducible shell/PowerShell entry points.

Adaptive V.35 Sampling Monitor | SystemVerilog, CDC, Python Independent Project | 2026

- Turned a field clock/data edge-coincidence failure into a measurable interface: designed synchronizers, event tracking, control state machines, CPU-visible result/status registers, timeout handling, and Python analysis; closed 29 self-checking system tests.

PROFESSIONAL EXPERIENCE

MetaX June 2026–September 2026

Board Design Intern Beijing, China

- Developed Python automation for GPU board validation and post-silicon characterization, converting high-speed interconnect measurements, repeated-test data, BOM revisions, and power/thermal logs into structured engineering results.
- Processed 61 Touchstone datasets with frequency extraction, interpolation, anomaly ranking, and report generation; built repeatability checks to flag abnormal variation for investigation.

ZGMicro July 2025–September 2025

RF Signal Processing Intern Remote

- Built MATLAB simulations for a BLE 2M PHY/GFSK receiver and analyzed digital down-conversion, filtering, synchronization, and fixed-point implementation tradeoffs.

Tiger Microwave August 2024–September 2024

FPGA Design Trainee Sichuan, China

- Implemented Moore/Mealy FSMs in Verilog with simulation testbenches while learning the FPGA design, synthesis, and implementation flow.